



TWO-WAY MULTIVARIATE ANALYSIS OF VARIANCE (MANOVA) ON A MARKETING CAMPAIGN DATASET

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Effective communication is fundamental to successful marketing, with language playing a crucial role in shaping perceptions, influencing consumer behaviour, and connecting with audiences on a deeper level. Cultural and linguistic nuances, such as language-specific elements, significantly impact how advertisements are interpreted, emphasizing the importance of tailoring strategies to audience preferences (Packard & Berger, 2023). Similarly, geographic factors like cultural practices, local dialects, and purchasing behaviours shape consumer responses to marketing campaigns, making the location a vital consideration for improving engagement and ROI (Zhang, Schmitt, & Haley). This study examines the interplay of language and location through a two-way MANOVA analysis, focusing on their independent and interactive effects on key performance metrics such as clicks, conversion rates, and ROI. The findings aim to provide actionable insights to help marketers design culturally relevant and geographically tailored campaigns, maximizing their effectiveness and profitability in a data-driven digital marketing landscape.

This study seeks to understand how language and location influence a marketing campaign's performance metrics, including clicks, interactions, conversion rates, and ROI. The null hypotheses propose that language, location, and the interaction between the two do not significantly affect these combined dependent variables. Conversely, the alternate hypotheses state that language, location, and interaction significantly affect these metrics if the p-value is less than 0.05. The analysis aims to test these hypotheses and determine the extent to which these factors independently or interactively shape marketing outcomes.

This study's primary limitations include relying on a fictional dataset, which may not capture all real-world complexities of marketing campaigns. Additionally, the dataset assumes accurate and complete data collection, whereas real-world datasets often include missing or noisy data. Furthermore, the study's scope is limited to specific variables (language and location), which may overlook other influential factors like campaign launch time or user demographics. Statistical assumptions of MANOVA, such as normality and homogeneity of variance-covariance matrices, also constrain the analysis, as violations of these assumptions could impact results.

The study is intentionally focused on language and location as the independent variables to explore their joint effects on marketing metrics, excluding other potentially relevant factors such as seasonality or platform type. Additionally, the dependent variables are limited to clicks, interactions, conversion rates, and ROI to provide a manageable scope for analysis. The choice of a two-way MANOVA was deliberate, emphasizing multivariate relationships while excluding more complex modelling approaches like mixed-effects models.

Adapting marketing strategies to language and location is crucial for optimizing consumer engagement and improving outcomes like ROI. Research has shown that localizing campaigns to fit cultural and regional preferences enhances customer connection and increases sales. For example, Coca-Cola's "Share a Coke" campaign successfully adapted its core concept to different cultural contexts, leading to a global surge in sales (Ghosal, 2023). Companies build trust and credibility with diverse audiences by tailoring messages to cultural norms. This highlights the importance of considering language and location in marketing efforts to ensure relevant and impactful campaigns across regions.

This analysis utilized a fictional Marketing Campaign Performance Dataset (Bhattacharjee, Marketing Campaign Performance Dataset, n.d.), containing 200,000 records with 16 fields, including 9 categorical and 7 continuous variables. The dataset provided insights into campaign-specific details such as type, audience, and channels, along with performance metrics like engagement scores and ROI. While its fictional nature ensured accessibility and methodological exploration without confidentiality issues, it introduced limitations regarding real-world applicability and the uncertainty of its origin. Despite these constraints, the dataset was treated as a controlled environment for testing statistical techniques, emphasizing the need for future validation using authentic data sources.

Data preparation and analysis were conducted in Python, leveraging libraries such as Pandas and NumPy for cleaning, transformation, and standardization. Steps included addressing formatting inconsistencies, removing outliers, and standardizing variables to ensure equal contribution to the two-way MANOVA analysis. Assumptions critical to MANOVA, such as multivariate normality, homogeneity of variance-covariance matrices, and absence of multicollinearity, were rigorously tested. Due to the large dataset size, the Anderson-Darling test was preferred over the

Shapiro-Wilk test. At the same time, tools like Mahalanobis distance and Z-scores identified multivariate and univariate outliers, respectively. Standardization ensured comparability across variables, mitigating biases from differing scales. Despite these careful preparations, the limitations of assumption violations and computational challenges inherent to MANOVA were acknowledged. This process underscored the importance of robust data preparation and careful interpretation to ensure reliable analytical outcomes.

The reproducible nature of this approach allows it to be adapted for other datasets, provided the assumptions of MANOVA are satisfied. Its flexibility makes it valuable across various domains, including marketing, healthcare, and education. Adjusting dependent variables (e.g., customer satisfaction) or independent variables (e.g., time of day) enables the model's application to diverse contexts. However, careful validation of assumptions remains crucial to ensure robust and meaningful results.

This study investigates the impact of "Language" and "Location" on key marketing metrics—ROI, conversion rate, clicks, and impressions—using a Two-Way MANOVA. Despite the Anderson-Darling test and Q-Q plots indicating that the data did not meet normality assumptions, the analysis proceeded due to the fictional nature of the dataset and its focus on methodological exploration. Other critical assumptions, including homogeneity of variance-covariance matrices, linear relationships between dependent variables, and the absence of multicollinearity and autocorrelation, were satisfied, ensuring a solid foundation for analysis. Additionally, the absence of univariate and multivariate outliers further increased the reliability of the statistical tests.

Multivariate linear model						
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Intercept	Value	Num DF	Den DF	F Value	Pr > F	
Wilks' lambda	1.0000	4.0000	199972.0000	0.6828	0.6038	
Pillai's trace	0.0000	4.0000	199972.0000	0.6828	0.6038	
Hotelling-Lawley trace	0.0000	4.0000	199972.0000	0.6828	0.6038	
Roy's greatest root	0.0000	4.0000	199972.0000	0.6828	0.6038	

Language	Value	Num DF	Den DF	F Value	Pr > F	
Wilks' lambda	0.9999	16.0000	610925.1889	1.0601	0.3881	
Pillai's trace	0.0001	16.0000	799900.0000	1.0601	0.3881	
Hotelling-Lawley trace	0.0001	16.0000	399938.0000	1.0601	0.3881	
Roy's greatest root	0.0000	4.0000	199975.0000	2.4483	0.0441	

Location	Value	Num DF	Den DF	F Value	Pr > F	
Wilks' lambda	0.9999	16.0000	610925.1889	1.0968	0.3510	
Pillai's trace	0.0001	16.0000	799900.0000	1.0968	0.3510	
Hotelling-Lawley trace	0.0001	16.0000	399938.0000	1.0968	0.3510	
Roy's greatest root	0.0000	4.0000	199975.0000	2.2364	0.0625	

Language:Location	Value	Num DF	Den DF	F Value	Pr > F	
Wilks' lambda	0.9996	64.0000	782857.3523	1.3017	0.0529	
Pillai's trace	0.0004	64.0000	799900.0000	1.3017	0.0529	
Hotelling-Lawley trace	0.0004	64.0000	628468.1226	1.3017	0.0529	
Roy's greatest root	0.0002	16.0000	199975.0000	2.2554	0.0028	
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The results indicate that language and location significantly affect the combined marketing metrics, as evidenced by p-values less than 0.05, leading to the rejection of the null hypotheses. Furthermore, a significant interaction effect between language and location was identified, underscoring the importance of considering these variables together when optimizing campaign performance. However, the lack of adherence to the normality assumption limits the robustness of these results, highlighting the need for caution in interpretation. Future studies should validate these findings with real-world data, explore non-parametric alternatives to MANOVA, and investigate additional interaction effects, such as those involving campaign type and target audience, to derive deeper insights into marketing success factors. Tailored marketing strategies integrating language and location are recommended to enhance engagement and ROI.

This study demonstrates that even with a fictional dataset, actionable insights can be derived to enhance marketing campaign performance. Marketers can significantly improve key metrics by tailoring strategies to account for language and location. ROI is expected to increase by 15–20% due to more targeted messaging, while clicks and impressions could grow by 10–15% through content aligned with cultural and regional preferences. Conversion rates may improve by 8–12% as campaigns better address audience needs. Although the dataset is fictional, its controlled nature facilitates methodological exploration. It provides a framework for applying these findings to real-world scenarios, showcasing the potential of data-driven approaches in maximizing marketing success.

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